

SUJAL ADHIKARI

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EDUCATION

Caldwell University

B.S. in Computer Science and Mathematics

Caldwell, New Jersey

Expected Graduation: May 2028

- **Concentrations:** AI/ML & Data Science | **GPA:** 3.85/4.00, **Dean's List, Honors Program**

EXPERIENCE

Undergraduate Student Researcher

C-STEM Laboratories, Caldwell University

Caldwell, New Jersey

May 2026 - Current

- Implemented **FedAvg**, **FedProx**, and **FedNova** from scratch in PyTorch on **CIC-IDS2017/2018** across four **non-IID silos**; found FedAvg/FedProx collapsed to ~50-58% accuracy on the minority silo while FedNova sustained 96-98%.
- Diagnosed **step-count normalization** as the critical aggregation mechanism under extreme silo-size imbalance; corrected FedNova averaged ~95% accuracy across all silos on both **CIC-IDS2017** and **CIC-IDS2018** (range: 87-99%), outperforming centralized training's instability on **minority attack classes**.
- Identified **minimum viable silo size** as a fundamental bottleneck in federated intrusion detection under realistic **attack-type partitioning**.

PUBLICATIONS

Adhikari, S., Paudel, S., et al. "*Federated Learning for Network Intrusion Detection under Non-IID Silo Partitioning.*" Manuscript completed, targeting IEEE BigData 2026.

PROJECTS

EventGateway, AWS (SQS, RDS, ECS), SQLAlchemy, FastAPI, Docker

[GitHub](#)

- **Engineered** an asynchronous event ingestion pipeline in Python using **FastAPI**, **AWS SQS**, and **PostgreSQL (RDS)**, decoupling request ingestion from data persistence via an independent worker consumer to keep API latency unaffected by downstream processing load.
- **Enforced** least-privilege access control across services with separate **IAM** task roles per container — the producer restricted to **sqs: SendMessage** with no database access path, and the worker scoped to **SQS** consume/delete plus **RDS** connectivity — isolating the blast radius of each component.
- **Containerized** the **producer** and **worker** as separate services with **Docker**, deployed on **AWS ECS Fargate** with dedicated task definitions, and validated end-to-end message flow from live API request through **SQS** to **RDS** persistence using **CloudWatch** log tracing.

Clarity — Volatility Forecasting & Model Benchmarking Pipeline, Web Scraping, MLOps, Automation

[GitHub](#)

- **Engineered** an **automated** end-to-end forecasting pipeline in Python, ingesting historical market data (**2008–present**) through the **Tiingo API** for 21 large-cap equities and S&P 500 benchmarking, with incremental data refresh and feature engineering including rolling volatility, EWMA, and log returns.
- **Benchmarked** four forecasting architectures — **XGBoost**, **LSTM**, **RNN**, and **GARCH** — on volatility prediction; XGBoost achieved the strongest performance ($R^2=0.75$), outperforming neural and econometric baselines.
- **Deployed** a containerized inference service using **Docker**, serving predictions via a **FastAPI REST API** on **Render**, with a **Streamlit** frontend on Streamlit Community Cloud for interactive model comparison and forecasting.

Hierarchical Network Intrusion Detection, Anomaly Detection, Feature Engineering

[GitHub](#)

- **Engineered** a multi-stage ML pipeline on **125K imbalanced NSL-KDD** records, stacking **Random Forest** and **XGBoost** after benchmarking baseline models.
- Achieved 99.4% **F1** with **false alarms** below 1%; improved **minority-class detection** on rare attack types to 97.4% F1 despite extreme data scarcity.
- Achieved 100% **precision/recall** on baseline **models** and 97.4% **F1** on rare attack types under **extreme data scarcity**.

ACTIVITIES

Outreach Lead, Google Developers Group at Caldwell University

Caldwell, New Jersey

- Grew Google Developers Group to 100+ members; organized AI/ML workshops, hackathons, and speaker events.

SKILLS

Languages & Libraries: Python, SQL, PostgreSQL, SQLAlchemy, NumPy, Pandas, SciPy, scikit-learn, XGBoost, PyTorch

ML & Data Science: Machine Learning, Deep Learning (CNN, LSTM, GRU), Federated Learning (FedAvg, FedProx, FedNova), Hugging Face, Statistical and Hypothesis Testing

Cloud & Deployment: AWS (SQS, RDS, ECS), Docker, Render, Streamlit Cloud

Tools: FastAPI, Streamlit, Git, GitHub, VS Code, Jupyter Notebook, MLOps, Hugging Face